How to setup endpoint authentication and network security in OData service framework

This article explains which are the steps to setup the network security and the different kinds of authentications to configure an OData service.

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Title
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Summary
This article explains which are the steps to setup the network security and the different kinds of authentications to configure an OData service.

From Version

Up to Version
5.3 SP21

Addressees

Context

Background Info
OData eases the access of data from external connectors without a direct connection to the database. Consumers can access CCH® Tagetik data directly via HTTPs requests. The security of the connection is fundamental and requires to properly setup some network components. Authentication type choice affects the creation of the users involved in the use of the OData service. CCH® Tagetik cloud team intervention is always required, Customer IT intervention is required in case of OAuth2 Authentication.

Procedure
The components involved in the configuration are:
- Endpoint and Authentication.
- Firewall and whitelist.

Endpoint creation

1. From Repository Homepage access the tile **Endpoint**.
2. Add a new endpoint and define code and description.
3. Select as Endpoint type **OData V4 service** or **OData V2 service** depending on which version the source exposes the OData service (e.g., SAP exposes its data in version 2).
4. Define the OData service URL.
5. Select in the combo box **Available authentication type** among None, Basic Authentication and OAuth 2.0 authentication.

Basic authentication

The user that must be used to authenticate is the local user defined in the paragraph **User's creation for Basic authentication** of the article OData: General settings and Users setup (/CustomerPortal/s/article/OData-General-settings-and-Users-setup)

1. Choose **Basic** in the check box **Available authentication**.
2. Type the code of the local Tagetik user in the text box **OData service username**.
3. Type the password of the local Tagetik user in the text box **OData service password**.

OAuth2 authentication

1. Choose **OAuth 2.0** in the check box **Available authentication**.
2. Open an INCIDENT on CCH® Tagetik support portal to Cloud team or IT in which the implementer will ask to create configuration file with the OpenID properties to be stored on the application server's local filesystem. These properties have the information that the CPM will use to communicate with the OpenID provider. Open an INCIDENT on CCH® Tagetik support portal to Cloud team to create configuration file with the OpenID properties to be stored on the application server's local filesystem
3. Customer IT has to configure, on OpenID provider, an OAuth 2.0 token with Client Credential Flow. Each Scope must be coded as the related Reference user defined in CCH® Tagetik. This token should contain one or more usernames of the Reference Users created above as Scopes (Configure an OAuth 2.0 token with Client Credential Flow on OpenID provider (Customer IT resposibility)).

OAuth2 is not supported directly by PowerBI. In case customer should create a specific driver to connect API to PowerBI. Tagetik does not support this implementation.

Firewall and whitelist

All IP Public addresses that need to connect to CCH® Tagetik and to OData, must be whitelisted due to proxy. To accomplish this step, it is necessary involve the CCH® Tagetik cloud team.
The implementer must:
1. Open an INCIDENT on CCH® Tagetik support portal to Cloud team that includes IP Public addresses that need to connect to CCH® Tagetik and to OData.
2. Provide the URL used to access the CCH® Tagetik application.

Cloud team will manage to update the whitelist and to setup properly the file standalone.xml.

IPs to whitelist in case of Power BI cloud-based or standalone

When PowerBI is installed on a local computer, the list of IPs to be whitelisted can be univoquely identified. Instead, when deployed on the cloud, the IP depends on the region and on the service required; if it is deployed on Microsoft Azure, Microsoft itself provides a range of available IP addresses, among which the Customer IT have to choose the ones to use. Here is the link to use to retrieve the MS Azure IPs: https://www.microsoft.com/en-us/download/details.aspx?id=56519 (https://www.microsoft.com/en-us/download/details.aspx?id=56519) . If PowerBI is deployed using a different provider, Customer IT should contact the provider to request the available IPs.

Example

Troubleshooting

Technical Requirement

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